

西南财经大学本科期末考试试题册（B）

（2025—2026 学年第 1 学期）

以下各项由命题教师填写：

课程名称：数据结构（英） 命题教师：陈中普、王海林

适用对象（年级专业）：2024 级金融数学、2024 级信息管理、2024 级物流管理

使用试题的任课教师姓名：陈中普、王海林

试题说明：

- 1、考试类型：闭卷☒ 开卷☐
- 2、本套试题共 四 道大题，共 7 页，完卷时间 120 分钟。
- 3、考试用品中除纸、笔、尺子外，可另带的用具有：
计算器☐ 字典☐ 等
(请在下划线上填上具体数字或内容，所选☐内打钩)

考试时间（由制卷方填写）：

以下各项由学生填写：

任课教师：

年级专业：

学生姓名：

学 号：

- 考生注意事项：
1. 出示学生证（或身份证）和准考证并放置于桌面左上角，以备查验。
 2. 领取试卷后清点并检查试卷页数，如有重页、页数不足、空白页及印刷模糊等举手向监考教师示意调换试卷。
 3. 答题前先将试题册及答题纸上的任课教师、专业、年级、学号、姓名等信息填写完整。
 4. 所有答案均需填写在答题纸上，填写在试题册上无效。
 5. 考生不得携带任何通讯工具进入考场。
 6. 考试结束后，将试题册、答题纸和草稿纸等材料上交监考教师，不得带离考场。
 7. 严格遵守考场纪律。

By default,

- The height of an empty tree is -1.
- A singly linked list has only its head node as an attribute.
- The node in a singly linked list has attribute data and next, in which data is the element it is holding, and next is the link to the next node.
- A binary tree has only its root node as an attribute.
- The node in a binary tree has attribute key, left, and right, in which key is the data it is holding, and left and right are links to its left and right child, respectively.

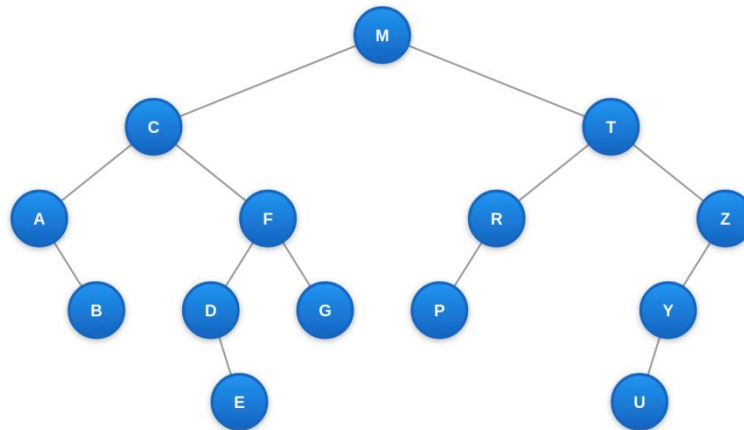
Part One: Multi-choices (2 marks each; 30 marks in total)

(Note: Each question has only one correct choice.)

1. Which of the following statements is NOT correct in terms of big O notation?
 - A. $f(N) = N^3 + 6N^2 + 3N\log N$, then $f(N) = N^3$.
 - B. If $f_1(N) = O(N\log N)$ and $f_2(N) = O(N\log N)$, then $f_1(N) = f_2(N)$.
 - C. A hash table not always guarantees $O(1)$ worst-case time for every search operation.
 - D. Consider a list in Python, 'Appending' the last element can be done in $O(1)$ when using amortized analysis.
2. Which pair of operations gives the worst-case $O(1)$ behavior for both enqueue and dequeue when a stack is built on a singly linked list with only a head pointer?
 - A. enqueue at tail, dequeue at tail.
 - B. enqueue at head, dequeue at head.
 - C. enqueue at tail, dequeue at head.
 - D. enqueue at head, dequeue at tail.
3. When using a array to build a circular queue, which index update formula correctly moves the front pointer one step forward after a dequeue operation?
 - A. $\text{front} = \text{front} + 1$.
 - B. $\text{front} = (\text{front} + 1) \% \text{capacity}$.
 - C. $\text{front} = (\text{front} + 1) \% (\text{size} + 1)$.
 - D. $\text{front} = (\text{front} - 1 + \text{capacity}) \% \text{capacity}$.
4. Given a singly linked list that only maintains a head pointer (no tail or size field), which operation can be performed in $O(1)$?
 - A. Append a new node to the end of the list.
 - B. Remove the last node from the list.
 - C. Remove the first node from the list.
 - D. Count the total number of nodes in the list.
5. In a binary search tree containing the numbers 1, 2, 3, 4, 5, 6, 7, 8, the root is 6, the left-subtree root is 3, and the right-subtree root is 8. What is the minimum number of key comparisons needed to search for 4?

- A. 3
- B. 4
- C. 5
- D. 6

6. Given the sub-tree rooted at T bellow, what is sequence of a post-order traversal?



- A. $B \rightarrow A \rightarrow C \rightarrow E \rightarrow D \rightarrow F$
- B. $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F$
- C. $T \rightarrow R \rightarrow Z \rightarrow Y \rightarrow U \rightarrow P$
- D. $P \rightarrow R \rightarrow U \rightarrow Y \rightarrow Z \rightarrow T$

7. What is the time complexity of the following function in terms of N?

```

def bar(lst):
    N = len(lst)
    s = 0
    i = 1
    while i <= N:
        s += lst[i-1]
        i *= 2
    return s
  
```

- A. $O(1)$
- B. $O(N)$
- C. $O(\log N)$
- D. $O(N^2)$

8. Insert 20, 15, 13, 24, 9, 16, 32, 21 in that order into an initially empty binary search tree using the standard binary search algorithms. In the resulting BST, what is the left child of the node with value 24?

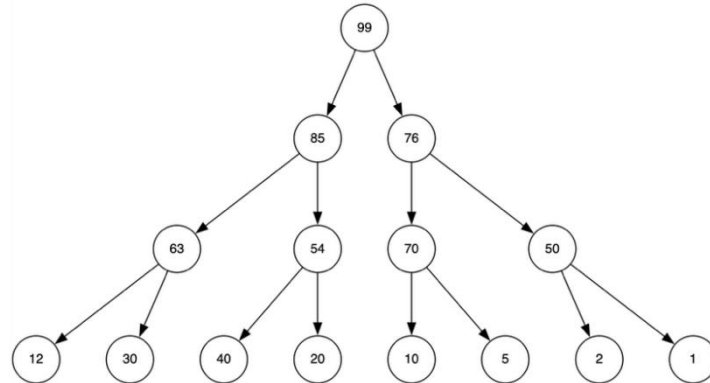
- A. 13
- B. 15
- C. 20
- D. 21

9. Which of the following statements is correct in terms of a red-black tree?

- A. A red node can have a red child.

- B. The root is always black.
- C. A leaf node (NIL) is counted as red.
- D. Search can be $O(N)$ in the worst case.

10. Given a max-heap below, after deleting the maximum 2 times using the standard heap algorithm, who is the right child of node 70?



- A. 50.
- B. 5.
- C. 10.
- D. 2.

11. For elements $\{1, 14, 4, 20, 8, 25, 36, 26, 5\}$, the hash function is $\text{hash}(x) = x \bmod 10$, and the linear probing is used for open addressing. Suppose elements are stored in an array with 10 slots (from slot 0 to slot 9), what is the element at slot 9?

- A. 5
- B. 25
- C. 26
- D. 36

12. Which of the statements is correct in terms of a hash table?

- A. Open addressing can always store more items than the table size.
- B. Separate chaining requires the load factor to be ≤ 1 .
- C. In open addressing, the load factor must be kept ≤ 1 .
- D. Hash collisions can be completely avoided by a perfect hash function in every application

13. For a connected, undirected graph with n vertices and m edges stored as an adjacency list, what is the time complexity of a single BFS traversal starting from a given vertex?

- A. $O(mn)$
- B. $O(m + n)$
- C. $O(m^2)$
- D. $O(n^2)$

14. Which of the following statement is correct in terms of a graph?

- A. DFS always gives the shortest path between any two vertices.
- B. BFS can be implemented with a stack instead of a queue.
- C. An undirected graph can have at most n^2 edges.

D. A tree with n vertices has exactly $n-1$ edges and is always connected.

15. While debugging a program, you want to check that every opening bracket “(”, “[”, or “{” has a matching closing bracket in the correct order. Which data structure lets you verify the brackets in $O(n)$ time?

- A. Stack
- B. Queue
- C. Hash table
- D. Tree

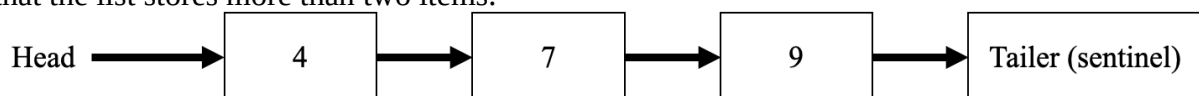
Part Two: Blank-filling (2.5 marks each; 25 marks in total)

1. The following pseudo code snippet is to put a node with key k into BST (i.e., bst) . Fill in the blanks in (1)-(4).

Algorithm insert(bst, k)

```
x ← bst.root
y ← null
z ← bst.Node(key)
while _____ (1) _____ do
  y ← x
  if _____ (2) _____ then
    x ← x.left
  end
  else if key > x.key then
    _____ (3) _____
  else
    return
end
if _____ (4) _____ then
  root ← z
else if key < y.key then
  y.left ← z
else
  y.right ← z
```

2. Suppose we use a tail-sentinel node (containing no real data) at the end of a singly linked list (slt). The list only keeps a pointer tailer that always points to this sentinel node, and assumes that the list stores more than two items.



Please Fill in blank (5) to complete the pseudo-code that removes the last real node (while the tail-sentinel must remain).

Algorithm remove_last(lst)

```
x ← slt.head
while _____ (5) _____ then
  x ← x.next
end
x.next = tail
```

3. The following pseudo code snippet is to find the maximum of nodes in a binary tree (i.e., tree) by calling $max(tree.root)$. Please fill the blanks in (6) and (7).

(Hint: It is a recursive algorithm)

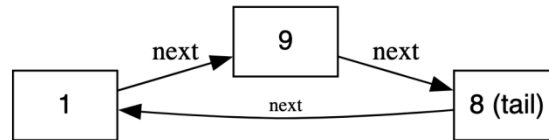
Algorithm max (node)

```

if node.right == NULL then
    _____ (6) _____.
end
return _____ (7) _____.
end

```

4. In a singly linked list, if the last node (i.e., *tail*) also has a reference to its first node (i.e., *head*), then it is called a *circularly linked list*. It is clear that it is sufficient to store only tail node in a circularly linked list. The following figure depicts $1 \rightarrow 9 \rightarrow 8$.



Please fill the blanks in (8) and (9) to search the minimum key of a circularly linked list (i.e., *slt*).

(Hint: *lst* has only the tail node, nothing more.)

Algorithm `get_min(slt)`

```

tail ← lst.tail
if tail == NULL then
    return 0
end
min_val ← tail.value
current ← tail.next
while _____ (8) _____ do
    if current.value < min_val then
        min_val ← current.value
    end
    _____ (9) _____
end
return min_val

```

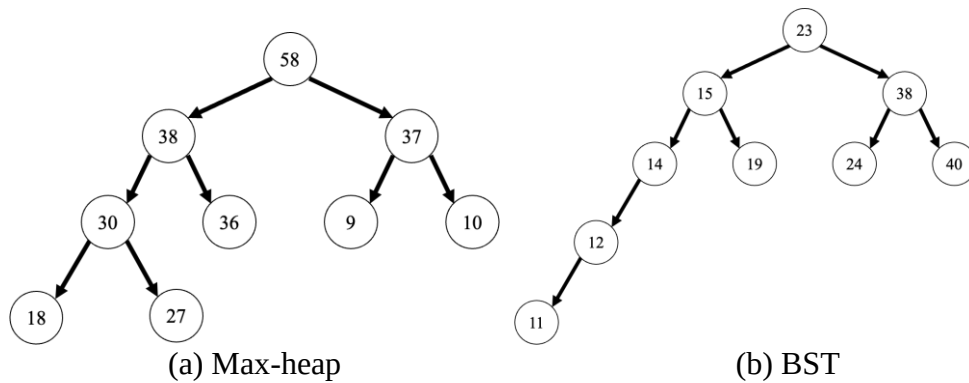
5. If the load factor is α in a separate-chaining hash table, the expected time complexity for a lookup is ____ (10) ____ time.

Part Three: Short Q&A (5 marks each; 10 marks in total)

1. A BST maintains the property that for every node, all keys in its left subtree are smaller and all keys in its right subtree are larger.

- Explain why searching for a key in a BST has a worst-case time complexity of $O(N)$, even though the average time is $O(\log N)$.
- What BST property ensures that the search operation can ignore half of the remaining subtree at each step?

2. Draw figures.



- Given the above max-heap (a), draw the resulting figure after inserting 42.
- Given the above BST (b), how to convert it into an AVL tree through rotation?

Part Four: Analysis and Design (10 + 10 + 15; 35 marks in total)

1. A string is called a palindrome, if it reads the same forward and backward. For example: "aba", "level", and "racecar" are palindromes. Write a recursive function that determines whether a given string *s* is a palindrome.

- If the length of *s* is 0 or 1, return True (it is a palindrome).
- Otherwise:
 - Compare the first and last characters.
 - If they are different, return False.
 - If they are the same, recursively check the substring obtained by removing the first and last characters.
- You must use recursion.

Do not use any built-in reverse operations.

(Hint: You can use pseudo-code, and even plain-English as long as it is clear enough.)

```
def is_palindrome(s: str) -> bool:
    # Your implementation here
```

2. Please prove an AVL tree with *N* nodes has height $O(\log N)$.

(Hint: We use $N(h)$ to denote the minimum number of nodes in an AVL tree with height *h*.)

3. In a university course registration system, each student has a singly linked list of courses they've enrolled in. Each course contains its code and semester. Each student is also represented as a node in a graph. There is an undirected edge between two students if they have taken at least one course together.

(Hint: You can use pseudo-code, and even plain-English as long as it is clear enough.)

(a) Each Course has attributes including *code*, *semester*. A course list has a *head* node, representing the first course.

```
class Course:
    def __init__(self, code: str, semester: str, next=None):
        self.code = code
        self.semester = semester
        self.next = next

class CourseList:
    def __init__(self, head=None):
        self.head = head
```

Given a CourseList object lst, implement an algorithm to count how many courses a student has taken in the “Fall 2025” semester. (5 marks)

```
def count_fall_2025_courses(lst: CourseList) -> int:  
    # Your implementation here
```

(b) Given a graph G where G.adj[i] returns a list of students who have shared a course with student i, and a mapping C[i] gives the course list of student i, implement an algorithm to compute the maximum number of courses among student i and all students reachable from i (directly or indirectly). (10 marks)

```
def max_courses_in_group(G, C, i) -> int:  
    # Your implementation here
```